A

Major Project Report

On

"AI – DRIVEN SMART SURVEILLANCE FOR WEAPON DETECTION USING YOLOv8"

Submitted in partial fulfillment of the
Requirements for the award of the degree

IN

MASTER OF COMPUTER APPLICATIONS

Submitted By

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ANURAG UNIVERSITY

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(2023 – 2025)

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CERTIFICATE

This is to certify that the project report entitled **"AI – Driven Smart Surveillance for Weapon Detection using YOLOv8"** is a Bonafide work done and submitted by **Kolloju Srinivas Bharath Raj (23MC201B46)** in partial fulfillment of the requirements for the award of the degree of MASTER OF COMPUTER APPLICATIONS **from Anurag University** during the academic year 2024 - 2025 and the Bonafide work has not been submitted elsewhere for the award of any other degree.

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ACKNOWLEDGEMENT

I would like to express my sincere thanks to **Dr. K. Sudheer Reddy**, Dean of Academics and Planning, Anurag University, Ghatkesar, whose motivation in the field of software development has made me overcome all hardships during the study and successful completion of the project.

I would like to express my profound sense of gratitude to all for having helped me in completing this dissertation. I would like to express my deep-felt gratitude and sincere thanks to my guide, **Dr. K. Nagamani**, Assistant Professor, Department of Information Technology, of Anurag University, Ghatkesar, for her skilful guidance, timely suggestions, and encouragement in completing this project.

I extend my sincere appreciation to **Dr. V. Vijaya Kumar**, Dean, School of Engineering, for his encouragement and consistent support during this endeavor.

A special acknowledgment goes to **Mr. G. Sekhar Reddy**, Program Coordinator, MCA whose support and guidance played a pivotal role in the successful completion of this project.

Finally, I would like to express my heartfelt thanks to my parents who were very supportive both financially and mentally and for their encouragement to achieve my set goals.

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DECLARATION

This is to certify that the major project titled **"AI – DRIVEN SMART SURVEILLANCE FOR WEAPON DETECTION USING YOLOv8"** is a Bonafide work done by me in fulfillment of the requirements for the award of the degree **MASTER OF COMPUTER APPLICATIONS** and submitted to **Anurag University, Hyderabad.** I also declare that this project is a result of my own effort and has not been copied or intimated from any source. Citations from any websites are mentioned in the bibliography. This work was not submitted earlier at any other university for the award of any degree.

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ABSTRACT

The rising firearm dangers as well as the need for improved and rapid surveillance systems create an expanding market demand. A real-time weapon detection system recommends integration between YOLOv8 object detection and Django web server technology with OpenCV-based image processing is done for scalable and reliable performance.

The surveillance system operates by taking continual video feeds from cameras while it runs deep learning algorithms to detect weapons in the present moment. When weapons appear in front of the system it triggers instant notifications that reach both web platforms and mobile devices so personnel can stay updated in real time.

The weapon detection system based on dataset with advanced computer vision capabilities elevates situational understanding and improves detection accuracy and response efficiency for both public and private security environments. This system helps establish proactive threat management combined with automated intelligence to improve environmental security.

The system design includes flexibility for seamless integration with existing security systems which includes notification APIs and automatic lockdown functions and smart alarms. The flexible nature of this technology enables it to update with future improvements while preserving its fundamental mission of speeding up and making firearm detection more reliable for public security purposes.

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LIST OF ABBREVIATIONS

1. AI - Artificial Intelligence
2. CCTV - Closed-Circuit Television
3. CNN - Convolutional Neural Network
4. GPU - Graphics Processing Unit
5. IEEE - Institute of Electrical and Electronics Engineers
6. ML - Machine Learning
7. RAM - Random Access Memory
8. YOLO - You Only Look Once
9. YOLOv3, YOLOv4, YOLOv5, YOLOv8 - Different versions of the YOLO object detection model
10. ACM - Association for Computing Machinery
11. OpenCV - Open Source Computer Vision Library
12. NMS – Non-Maximum Suppression
13. R-CNN – Region-based Convolutional Neural Network
14. Fast R-CNN – A faster version of R-CNN
15. Faster R-CNN – With Region Proposal Network (RPN)
16. RPN – Region Proposal Network
17. SMTP – Simple Mail Transfer Protocol (used for Gmail alerts)
18. FPS – Frames Per Second
19. RGB – Red Green Blue color model
20. CV – Computer Vision (sometimes abbreviated in comments modules)
21. ROI – Region of Interest (typically used with bounding boxes)
22. IoU – Intersection over Union (for bounding box accuracy)
23. HTTP – HyperText Transfer Protocol
24. REST API – Representational State Transfer API
25. JSON – JavaScript Object Notation (used in data exchange)
26. IP – Internet Protocol

1. INTRODUCTION

Public space safety concerns are rising, and weapons-related threats become increasingly common. Traditional surveillance systems used by security authorities and law enforcement agencies, whether hidden or worn, are often difficult to recognize weapons. These traditional methods are based on human operators who monitor video surveillance materials. This is a process susceptible to inefficiencies, delays and errors that hinder timely threat responses.

The growing need for improved security has led to the development of automated monitoring solutions that integrate artificial intelligence (AI) and deep learning technologies to improve the accuracy of recognition and response speeds. In particular, real-time detection of objects recorded significant advances, allowing us to quickly analyze large amounts of visual data.

Automated monitoring solutions that utilize both artificial intelligence (AI) and deep learning technologies have developed because of increasing security requirements to enhance recognition precision alongside response times. Real-time detection of objects demonstrated remarkable progress which enables efficient processing of extensive visual data. The proposed research develops a Real-Time Weapon Detection System that integrates YOLOv8 along with OpenCV image processing and a Django-based backend storage. The system operates by processing current video frames to detect weapons immediately. The system uses a rapid notification system to instantly warn security staff who can then respond without delay to potential menacing situations.

The system reaches enhanced public security through deep learning combined with technology that enables quicker law enforcement response. The proposed solution delivers three main advantages for the system:

Security personnel can respond rapidly to threats after receiving automatic alerts through the system.

1.1 OVERVIEW OF THE DOMAIN CHOSEN:

The realm of computer vision has undergone major transformations with the introduction of deep learning, allowing machines to interpret and analyze visual data with unprecedented accuracy. Object detection, a critical application in computer vision, is often used in safety, monitoring and autonomous systems. Under various recognition algorithms, Yolo (you only see once) is one of the fastest and most efficient frames of real-time recognition tasks. The latest version of the Yolo family,

Yolov8, offers significant improvements in detection speed, accuracy and arithmetic efficiency. This makes it very suitable for weapon detection in real surveillance scenarios. By processing video frames in real time, Yolov8 allows security systems to quickly identify threats.

The integration of Django, similarly complements the abilities of surveillance systems. With secure, scalable storage, weapon detection systems can store, retrieve, and control massive quantities of visible data efficiently. The combination of deep learning, computer vision, infrastructure creates a strong framework for current safety applications, making sure public protection thru proactive chance detection.

1.2 OBJECTIVE:

The overall goal of this project is to create an intelligent real-time system for weapon detection that increases public safety and the efficiency of law enforcement responses. The rationale for focusing in this area comes from the enhanced security threats that are brought about by openly carried and hidden weapons in public areas, including airports, schools, shopping malls, and government buildings. Traditional surveillance systems usually depend on human observation, which is susceptible to fatigue, latency, and error, rendering them useless in stopping weapon-related threats.

1.2.1 Why choose this domain ?

a.Rising Security Concerns:

Weapon-related offenses have surged worldwide, which has turned public security into a fundamental problem. The traditional security methods of surveillance cameras and manual checks do not include instant threat identification features. The development of real-time weapon detection capabilities in automated systems proves to be effective in addressing the rising problem.

b.Limitations of Existing Surveillance Systems:

- Continuous human supervision of conventional CCTV-based systems leads to delayed responses due to human oversight, and sometimes results in false negatives.
- Security systems that do not take advantage of deep learning detection capabilities perform poorly in their threat identification function.
- Traditional surveillance installations fail to provide live monitoring alerts, allowing dangerous situations to develop beyond the reaction times of security personnel.

c.Advancements in Deep Learning and Computer Vision:

The advancements in artificial intelligence and deep learning technology have led to remarkable object identification breakthroughs through YOLO (You Only Look Once) models. Real-time surveillance demands an ideal detection technology which YOLOv8 provides through its high speed performance alongside precise detection and outstanding processing efficiency.

d.Enhancing Law Enforcement and Public Safety:

- Instant alerting mechanism helps security personnel to quickly respond to potential threats and so reduce the risk of mass violence or casualties.
- The technology, which automates the detection process, also reduces the margin for human error, improving security monitoring accuracy.
- The project is an inexpensive solution compared to costly labor-intensive security operations.

1.3 PROBLEM FORMULATION:

Weapons threats are a key danger to public safety in today's security environments, and also warrant advanced surveillance mechanisms. Conventional security methods like CCTV surveillance and manual inspection depend on human observations which can lead to mistakes, inefficiencies and delays. Such limitations prevent real-time threat detection and interference and raise the probability of violent actions.

Challenges in the AI-driven Development of Weapons Recognition Systems:

- Processing video frames on the fly for recognizing weapons.
- Minimizing false positives and false negatives to enhance the precision of recognition.
- Real-time alerting system to alert security guards

To evaluate the effectiveness of the proposed system, the following rating metrics are utilized:

1. **Accuracy:** Proportion of correctly classified instances over all predictions.

$$\textbf{\textit{Accuracy}} = \frac{\mathbf{TP+TN}}{\mathbf{TP+TN+FP+FN}}$$

2. **Precision:** Measures how many of the detected weapons were actual weapons.

$$\textbf{\textit{Precision}} = \frac{\mathbf{TP}}{\mathbf{TP+FP}}$$

3. **Recall (Sensitivity):** Ability to correctly detect weapons.

$$\textbf{\textit{Recall}} = \frac{\mathbf{TP}}{\mathbf{TP+FN}}$$

4. **F1-Score:** Harmonic mean between precision and recall to bring balance between both metrics.

$$\textbf{\textit{F1-Score}} = 2 * \frac{\mathbf{Precision*Recall}}{\mathbf{Precision+Recall}}$$

Where:

- True Positive (TP): Correctly detected weapons.
- True Negative (TN): Correctly identified non-weapons.

- False Positive (FP): Incorrectly identified weapons.
- False Negative (FN): Missed weapon detections.

1.3.1 Problem Solution:

To address these challenges, this study develops an integrated AI-related real-time weapon recognition system.

•Yolov8 for quick and accurate object recognition.

•Opencv for image processing and analysis.

•Django backend. It is organized for scalable provisioning.

An automatic alarm mechanism that immediately notifies security personnel when a weapon is recognized.

This system combines traditional security technology with classic technology for the latest technology for monitoring AI controls to violate the vertical boundaries of modern AI control monitoring systems.

1.4 SCOPE:

The full range of this research covers the development, implementation, and assessment of a real time weapons detection systems by using deep learning. It is intended to enhance security surveillance by computerizing weapon identification, warning creation, and safeguarded data storage, more quickly response event and exact menace discovery.

Key Areas Covered:

1. **Real-Time Weapon Detection:**

- Implementation of YOLOv8, a cutting-edge object detection model, identifies real-time video of current weapons with high accuracy. The implementation identifies real-time video of current weapons with high accuracy.
- Integration with OpenCV for image processing enhances identification efficiency.

2. **Automated Threat Alert System:**

- Immediate notification to security personnel is triggered when they recognize weapons, allowing for rapid response and reduction of threats.
- Potential integration with law enforcement databases for additional security measures.

3. **Scalability and Adaptability:**

- Deployment of a Django-based backend enables scalability and accessibility from a couple of locations.
- Customization options to adapt the system for specific security environments, including public spaces, airports, schools, malls, and government buildings.

4. **Model Optimization and Accuracy Enhancement:**

- Continuous development of the YOLOv8 version through training on a diverse dataset to reduce false positives and false negatives.
- Enhancing system performance below various environmental conditions, including low visibility, crowded areas, and different lighting conditions.

5. **Potential Future Enhancements:**

- Integration with facial recognition and behavior analysis for a more comprehensive security solution.
- Development of cellular packages for on-the-go security tracking and response.

1.5 LIMITATIONS:

Although the suggested RTWD system improves surveillance and security response, it has several limitations that should betakes into consideration.

1. **Environmental Factors:**

- The detection efficiency can drop in low-light conditions or with poor camera resolution.
- Real-time performance in outdoor environments may be compromised by extreme weather such as fog, rain or glare.
- Overlapping objects because of background clutter or crowded environments may lead to increased misidentifications.

2. **Dependency on Camera Angles and Frame Rates:**

- The system's performance is also impacted by the placement and quality of surveillance cameras.
- Motion-blur detection is common in fast-moving objects and low camera frame rates.

3. System Resource and Latency Issues:

- Because real-time processing consumes a lot of processing capability, it can be infeasible for low-end devices.
- It may need to be optimized for real-time performance when deploying the system to high-traffic surveillance applications.

4. Security and Privacy Concerns:

- The system does not conduct facial recognition, meaning it can't detect who is carrying a weapon, only that there is a weapon on premises.

5. Limited Scope of Threat Detection:

- The system is designed specifically for firearm detection and may not recognize knives, explosives, or other concealed weapons unless explicitly trained for them.
- Behavioral analysis and anomaly detection (e.g., identifying aggressive movements) are not included in the current implementation.

1.6 SYSTEM REQUIREMENTS:

1.6.1 Software Requirements:

- Programming Language: Python
- Deep Learning Framework: YOLOv8
- Image Processing: OpenCV
- Backend Framework: Django
- Development Tools: VS Code

1.6.2 Hardware Requirements:

- Processor: Intel Core i5 / AMD Ryzen 5 (i7 recommended)
- GPU: NVIDIA GTX 1650 or higher (RTX 3060 for better performance)
- RAM: 8GB (16GB recommended for multiple streams)
- Storage: 512GB SSD
- Camera: 1080p resolution, 30 FPS
- Network: 10 Mbps upload speed

2. LITERATURE SURVEY

2.1 OBJECTIVE OF THE LITERATURE SURVEY:

Our objective is to evaluate modern research about AI-driven weapon detection systems while studying various research approaches together with their technical limitations. By understanding the current research we can spot missing information and learn more about our proposal that unites YOLOv8 with OpenCV and Django functions.

2.2 IMPORTANCE:

Surveillance systems that use intelligent technologies drive researchers to focus on weapon detection through deep learning methods because of heightened market demand. YOLO-based object detection gets used in existing methods yet these methods typically fall short when it comes to real-time alert systems and as well as scalability features. The survey follows this organization:

- Research investigators studied preselected papers which focused on weapon detection.
- The examination deals with multiple methodologies while analyzing different datasets and related findings.
- Different approaches face various strengths and weaknesses when analyzed.
- The proposal acknowledges and focuses on unmet research needs which the new system intends to resolve.

2.3 LITERATURE REVIEW:

Multiple studies have demonstrated how deep learning models from YOLO (You Only Look Once) family detect weapons using artificial intelligence. Investigators have demonstrated multiple methods for weapon detection within real-time video feeds through various available datasets. The proposed solution faces ongoing restrictions because it does not provide real-time alerts nor supports storage and confronts scalability challenges.

1. Application of Deep Learning for Weapons Detection in Surveillance Videos

- Authors: Tufail Sajjad Shah Hashmi et al.
- Model Used: YOLOv3 & YOLOv4
- Dataset: 7,800 manually annotated images

Key Contributions:

- Compared YOLOv3 and YOLOv4 for weapon detection.
- The authors established a data collection which included manual annotations for 7,800 images.
- Cardiovascular accuracy combined with increased operational velocity stood out as the main advantages of YOLOv4.

Limitations:

- No real-time alert system.
- No storage solution.

2. Real-time Weapon Detection using Drone

- Authors: Deepak R. Hawale, Pavin S. Game
- Model Used: YOLOv3 with OpenCV
- Dataset: Custom dataset with 2,000 weapon images

Key Contributions:

- The system performed weapon detection in real-time by operating drone-mounted cameras managed by Raspberry Pi devices.
- The system processed video frames acquired from the computer through its processing operations.
- Alerts generated upon weapon detection.

Limitations:

- Internet dependency affects reliability.
- The low processing capabilities of Raspberry Pi affect the system's operational speed.
- The system omits available storage solutions which could be used to save detected frames.

3. Weapon Detection from Images using YOLO and OpenCV

- Authors: Divyanshi Chitravanshi et al.
- Model Used: YOLOv5 with Transfer Learning
- The private dataset contains multiple images of guns which served as the main collection for evaluation purposes.

Key Contributions:

- The use of transfer learning techniques helped to boost detection precision levels.
- The required processing power needed by the model to operate in real-time video applications was lower.

Limitations:

- No storage solution.
- Lacks an integrated alert system for real-time security response.

Research demonstrates the advancement of AI weapon detection technology while showing that the systems still require further development. The practical use of these systems faces restrictions because they do not have real-time alert systems nor scalability features which would be essential for security and law enforcement applications. A proposed system links YOLOv8 and Django to create fast reliable detection systems which provide real-time alerts for public safety improvement.

2.4 RESEARCH GAPS:

Research papers have demonstrated effective AI-based weapon detection methods although their approaches still present specific disadvantages among them.

- Security delay occurs because real-time alerts are absent which prevents prompt security responses.
- The system lacks storage functions for frame storage which results in inefficient storage and retrieval of detected images.
- The use of Raspberry Pi devices as well as computational performance restrictions become obstacles during model execution.
- Because of low representation of datasets across various domains the detection system shows reduced accuracy across different environmental conditions.

2.4.1 Addressing the gaps:

Our proposed Real-Time Weapon Detection System addresses these research gaps by:

- Implementing YOLOv8 for improved speed and accuracy.
- Integrating real-time alert systems for immediate security response.

This system improves weapon detection efficiency while ensuring scalability and rapid response capabilities.

2.5 PROJECT OBJECTIVES:

Research demonstrates the advancement of AI weapon detection technology while showing that the systems still require further development. The practical use of these systems faces restrictions because they do not have real-time alert systems nor storage or scalability features which would be essential for security and law enforcement applications.

This research seeks to accomplish the following three main tasks:

1.The focus of this work is to build an accurate weapon detection system which operates in real-time.

- The project will deploy YOLOv8 to deliver quick and high-performance object detection solutions.
- Dynamic image processing and real-time video examination occur through the OpenCV library.

2.Enabling instant security alerts upon weapon detection

- An automatic alert system will notify security personnel through implementation.
- Implementation of systems that lower the time needed to detect potential threats.

3.Reported analysis of YOLO versions for identifying peak operational potential

- A performance evaluation between YOLOv8 and past YOLO versions focuses on their effectiveness combined with processing rates and operational efficiency.
- The project needs to identify its optimal deployable configuration for real-time execution.

4.Security model performance will be strengthened through advanced processes paired with diverse datasets.

- The model's effectiveness can be improved through enhanced dataset quality which decreases wrong positive and negative detections.
- The model's performance benefits from training with diverse environmental conditions to achieve better robustness.

5.The system needs to adjust to changing security operational needs

- A system with adjustable features must be designed to match different surveillance requirements.
- Supporting integration with multiple security infrastructures.

2.6 PROJECT JUSTIFICATION:

The desired destinations resolve all inquire about holes uncovered all through the writing survey.

- Real-time location confinements vanish since YOLOv8 accomplishes expedient handling which makes a difference weapons distinguishing proof happen right away.
- The venture incorporates real-time caution usefulness which sets up programmed security notices not at all like comparable ponders in this field.
- By moving forward both datasets extend and demonstrate quality the framework recognizes veritable weapons more proficiently whereas blocking unnecessary wrong cautions.
- The venture brings a versatile arrangement that expands its usefulness to cover both airplane terminals as well as schools and law authorization organizations.

This framework fulfills its goals to progress AI-driven shrewd reconnaissance by utilizing profound learning and advances to progress open security nearby law authorization operational productivity.

3. METHODOLOGY

3.1 INTRODUCTION:

The research methods developed a real-time weapon detection system which combines both efficiency and scalability features. The combination of YOLOv8 and OpenCV with Django produces an accurate rapid solution that can be used for real-world security monitoring applications. This research adopts the following methodology structure:

- Proposed Methodology: Overview of the theoretical approach and system components.
- The proposed algorithm contains its explanation together with pseudocode details of implementation.
- The working process details appear in a flowchart together with a dataflow diagram as part of the methodology design section.

The chosen methodology outperforms traditional detection methods due to its enhanced speed with high accuracy alongside real-time operation for applications that need automated surveillance and security threat detection.

3.2 PROPOSED METHODOLOGY:

The system includes different essential components to function.

1. Dataset Preparation:

- The YOLOv8 solution receives training based on over 200 weapon images which have diverse labels in the dataset.
- The image annotation process follows manual participation so detection accuracy remains high.

2. Preprocessing of Data:

- The detection performance gets enhanced through OpenCV while the images go through normalization and dimension resizing as preprocessing steps.
- YOLOv8 utilizes augmentation training methods to manage diverse lighting conditions and how objects are concealed or located in different backgrounds.

3. Real-Time Video Capture:

- Alive CCTV camera video feeds run continuously for real-time observation processes.
- Extracted frames enter the YOLOv8 model which runs the processing system.

4. Weapon Detection using YOLOv8:

- The YOLOv8 model uses each frame to identify weapon elements.
- The detection of weapons leads to storage of the visual box along with confidence score and classification information.

5. Alert Mechanism:

- The system triggers an instant warning signal after weapon recognition occurs.
- The system triggers security personnel notifications through web alerts in unison with mobile projectile message alerts.

6. Web-Based Monitoring System:

- Through Django-based web interface users gain access to real-time system monitoring functions.
- Users can access:
  - Live video feeds
  - Detected weapon logs
  - System alerts and notifications

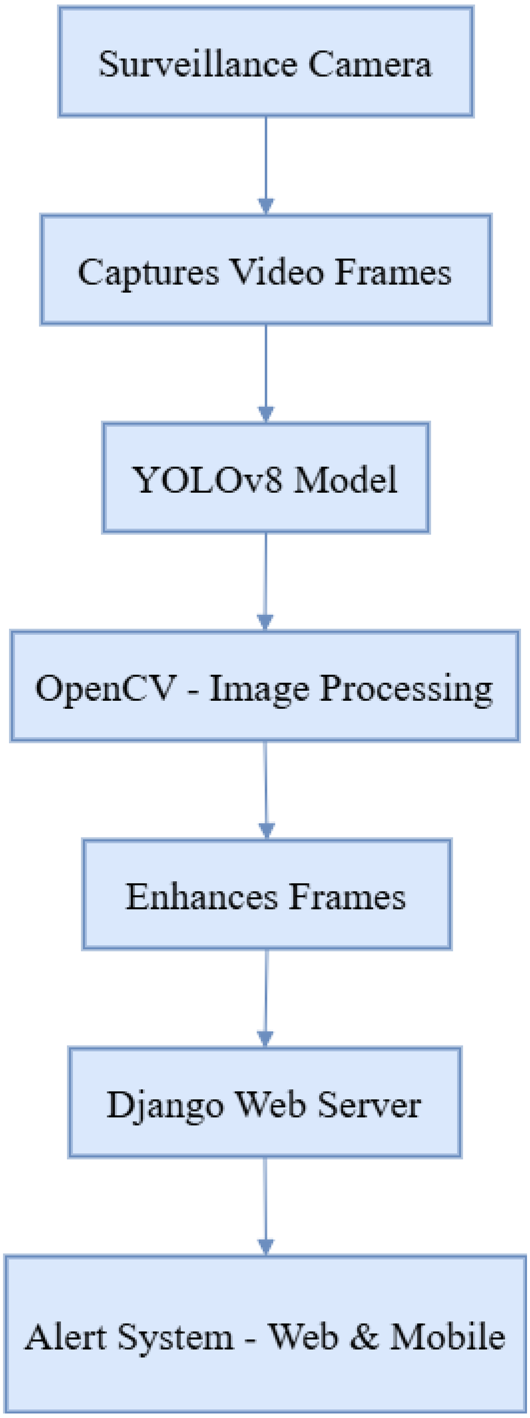


Fig 3.2. Proposed Methodology

3.3 PROPOSED ALGORITHMS:

The system depends on YOLOv8 model object detection technology alongside OpenCV implementation for image processing and Django platform for alert management and data storage. The following details every algorithm utilized from the beginning to the end of the pipeline:

```
BEGIN

1. Initialize System
   - Load YOLOv8 trained model
   - Connect to webcam
   - Load environment variables (for email, SMS)

2. User Authentication (Frontend)
   - IF user clicks "Register"
       Prompt user to enter details and store in database
   - ELSE IF user clicks "Login"
       Authenticate user from database

3. Enter Monitoring Details
   - Prompt user to input:
       - Area
       - Email or Mobile Number
   - Store details for alerting

4. Start Monitoring
   - Open webcam feed
   - Display Detection Window
5. Detection Loop
   WHILE monitoring is ON
       - Capture frame from webcam
       - Preprocess frame using OpenCV (resize, normalize)
       - Pass frame to YOLOv8 model
       - IF weapon is detected:
           - Draw bounding boxes on frame
           - Save frame locally

           - Send Email Alert (SMTP)
           - Send SMS Alert (ClickSend API)
           - Log detection to database
           - Display detection alert on GUI
       - ELSE:
           - Continue loop

6. Stop Monitoring
   - Close webcam
   - Return to main screen
7. View Detection Logs
   - Fetch logs from database
   - Display image, time, location, and alert status
END
```

3.3.1 YOLOv8 Algorithm:

The system implementation defines its full name as You Only Look Once Version 8.

Type: One-stage object detection algorithm.

Key Features:

- The object detection system employs one CNN to both perform classification and execute box regression tasks.
- The detection method employs anchor-free detection coupled with Decoupled Head and C2f (Cross Stage Partial) blocks as its core features.
- The algorithm uses CIoU Loss (Complete Intersection over Union) for enhancing box regression performance.

Detection Flow:

- The backbone of this network consists of CSPDarknet or a custom CNN to extract features.
- Path Aggregation Network (PANet) optimizes the features present in the feature maps through its operation in the neck stage.
- The prediction phase of the Decoupled detection head demonstrates both bounding box regression and class probability computation.
- The procedure yields (class_id, confidence, bbox coordinates) detections for each identified object.

Advantages:

- Real-time performance on edge devices.
- High accuracy for small object detection (like weapons).

3.3.2 Image Preprocessing Algorithms:

Normalize frame inputs: (used before detection)

- Bilinear Interpolation:
  - Resizes the frame to the size of the input model (e.g. 640×640).
- Min-Max Normalization:
  - Pixel values normalized to range [0, 1] using:

$$x_{\{norm\}} = \backslash frac\{x - x_{\{min\}}\}\{x_{\{max\}} - x_{\{min\}}\}$$

3.3.3 Non-Maximum Suppression (NMS):

- Goal: remove redundant bounding boxes for one object
- How it works:
  - Deduplicate detections using confidence score.
  - Retain the detection with maximum score
  - Discard other boxes with IoU (Intersection over Union) > threshold (e.g. 0.5).
- Used in: Post-processing of YOLOv8 predictions.

3.3.4 Intersection Over Union (IoU):

- The system uses this metric to determine predicted box overlaps with genuine labels.

Formula: $$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union}$$

- During training and evaluation of YOLOv8 the software deploys mAP@0.5 accuracy metrics.

3.3.5 SMTP Protocol (Email Notifications):

- Full Name: Simple Mail Transfer Protocol
- The Django server utilizes this protocol to notify users through Gmail SMTP emails.
- Authentication: Uses TLS encryption and app-specific passwords or OAuth2.

3.3.6 REST API (for Client-Server Communication):

- Protocol: HTTP-based client-server interaction
- Format: JSON payloads
- The REST endpoint of the Django server provides three main functionalities through the following interface points:
  - Sending alerts

- Storing metadata
- Accessing logs

3.3.7 ClickSend API (SMS Alerts):

- Type: HTTP-based SMS delivery service
- Working:
  - The application sends a POST request containing phone number and message data to ClickSend API.
  - Authentication via API key.

3.4 METHODOLOGY DESIGN:

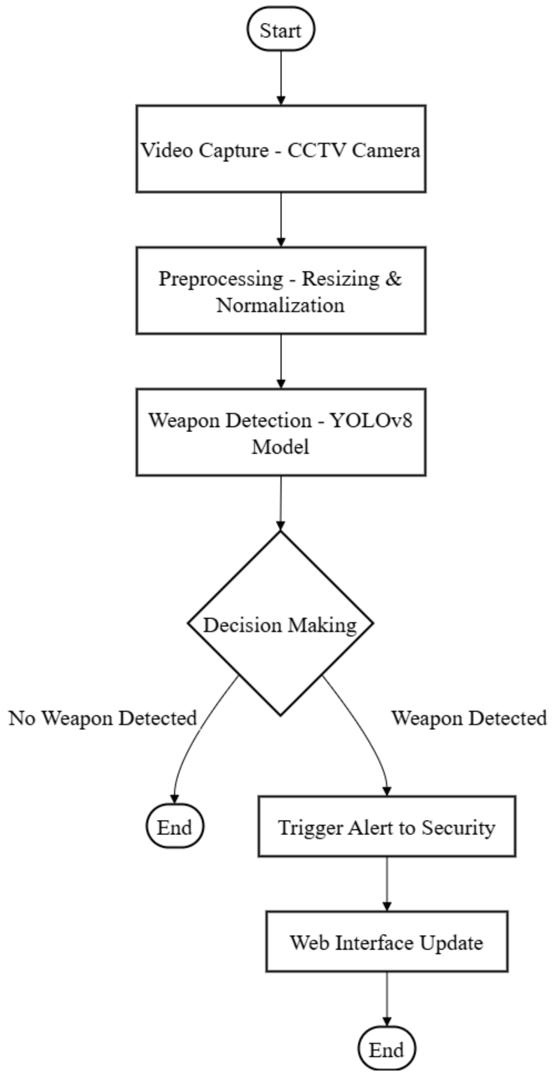


Fig 3.4. Methodology Design Flowchart

3.4.1 Flowchart Explanation:

1. Video Capture:
    - The system takes a live feed of video recording from installed CCTV cameras.
2. Preprocessing:
    - The system performs resizing and normalization and enhancement of frames via OpenCV.
3. Weapon Detection (YOLOv8 Model):
    - YOLOv8 is applied to all the frames that traverse through the system.
    - During the detection system process, it classifies the discovered objects as a weapon or a non-weapon.
4. Decision Processing:
    - If a weapon is detected:
        - Notify personnel of a security warning.
    - Else: Process video as usual.
5. User Interface:
    - Through the Django web application, security staff members access real-time detections and log files, along with security alerts.

4. IMPLEMENTATION

4.1 INTRODUCTION:

The Weapon Detection System follows an entire sequence for implementation as described in this chapter. The system completes the entire development process from creating datasets to preprocessing them and YOLOv8 model training before using OpenCV to integrate the model with live feeds, save data outcomes and send speedy warnings to protective staff.

4.2 DATASET AND PREPROCESSING:

Data Source and Collection:

Due to the lack of suitable existing public datasets the project built its own custom data collection. The collection of handgun images involved smartphone and webcam capture from multiple lighting conditions through different angles against various backgrounds for improving generalization capability. The data collection contains more than 200 snapshots with annotation labels.

Annotation:

The captured images were loaded into Roboflow because it serves as a popular annotation platform and dataset management system. The annotation process on Roboflow required users to draw boxes around handgun targets while providing necessary class assignments. The YOLOv8-compatible format became the final output of the exported dataset.

Preprocessing Steps:

- All images received standardized dimensions of 640x640 pixels because the YOLOv8 model requires such standardized input measurements.
- The pixel values received normalization treatment to establish consistent contrast and brightness levels which stabilized training operations.
- **Data Augmentation:**
  - Applied horizontal/vertical flips..
  - Random rotations and scaling.
  - A combination of brightness modification with contrast regulation and noise adjustment work. These modifications happened inside Roboflow and OpenCV framework to expand the initial dataset through artificial

methods. The team applied these modifications through Roboflow and OpenCV inside the workflow to artificially grow the dataset before helping the model identify variability in actual use cases.

4.3 IMPLEMENTATION:

Model Used:

Real-time handgun detection relied on the object detection architecture YOLOv8 (You Only Look Once version 8). Due to its combination of speed and precision YOLOv8 serves as the ideal security system for rapid and accurate purpose detection.

Training Configuration:

- Framework: Ultralytics YOLOv8
- Environment: Python 3.11.11, Torch 2.6.0 with CUDA 12.4 (Tesla T4 GPU)
- Training Set: 200+ annotated images
- Epochs: As per convergence
- Weights Path: runs/detect/train/weights/best.pt

Results:

- Precision: 0.998
- Recall: 1.000
- mAP@0.5: 0.995
- mAP@0.5:0.95: 0.965

The high detection accuracy along with low detection errors and missed alarms demonstrates the system quality.

Comparison with Existing Models:

YOLOv8 delivers enhanced precision and recall detection performance across dynamic environments only behind its instant ~2ms per frame inference speed.

4.4 EXPERIMENTAL SETUP:

The implementation was carried out in the following structured phases:

Phase 1: Dataset Preparation

- Manual image capture of handgun poses.
- Annotation using Roboflow.
- Export dataset in YOLO format.

Phase 2: Model Training

- YOLOv8 models received training within Google Colab platform by utilizing GPU hardware as an acceleration tool..
- Track metrics such as:
  - Precision (ratio of correct detections)
  - The recall percentage represented the number of real weapons detected out of all actual weapons present in the scene.
  - mAP (mean Average Precision across classes)

Phase 3: Integration with Real-Time Feed

- Access webcam/CCTV frames using OpenCV.
- Run YOLOv8 inference on each frame.
- The program displays both bounding boxes with annotations on top of each other.

Phase 4: Image Storage

- Convert detected frame to JPEG.

Phase 5: Alert Mechanism

- The detection image gets attached to an email that the system creates through Gmail SMTP server interface..
- ClickSend enables SMS transmission that includes both the image URL and detection data.

Phase 6: Web Interface

- The Django system processes requests at /upload_alert/ before it creates an image storage entry while creating a system log entry.
- A simple frontend shows:
  - List of detections
  - Live monitoring panel
  - Access to past alerts

5. PERFORMANCE EVALUATION

5.1 INTRODUCTION:

A weapon detection system requires superior performance features when used in real-time surveillance because speed and accuracy become essential factors. The system must recognize threats including handguns accurately so it generates minimum incorrect results and incorrect rejections. This chapter uses standard object detection metrics for performance assessment of the proposed system while offering results analysis through experimental outcomes.

5.1.1 Objective:

The evaluation of YOLOv8 after training requires testing its ability to detect handguns while assessing performance through accuracy measurements and reliability standards and speed capabilities. The system will generate rapid real-time response data at high accuracy levels for prompt alert notification.

5.2 PERFORMANCE METRICS USED:

The following metrics were used to evaluate model performance.

5.2.1 Precision:

The precision metric demonstrates the quantity of actual weapon detections that match the predictions among all predicted values.

$$\textit{Precision} = \frac{\mathbf{TP}}{\mathbf{TP+FP}}$$

Where:

- The correct identification of a handgun as a detected object stands as a True Positive (TP) result.
- The system delivered wrong handgun identifications as part of its incorrect detections.

Result: 0.998 – High precision implies very few false alarms.

5.2.2 Recall:

Model recall represents the functionality of identifying every existing weapon.

$$\textbf{\textit{Recall}} = \frac{\mathbf{TP}}{\mathbf{TP+FN}}$$

Where:

- FN (False Negative): Missed handgun in the frame.
- The test dataset contained no weapon detection errors because the achieved precision value reached 1.000.

5.2.3 mAP (Mean Average Precision):

Mean Average Precision (mAP) computes precision and recall rates by evaluating IoU thresholds between 0.5 to 0.95.

- mAP@0.5: Measures average precision at IoU threshold of 0.5
- mAP@0.5:0.95: Average across thresholds from 0.5 to 0.95 (step of 0.05)

$$\boldsymbol{mAp} = \frac{\mathbf{1}}{\mathbf{N}}\sum_{\boldsymbol{i=1}}^{\boldsymbol{N}} \boldsymbol{APi}$$

Results:

- mAP@0.5 = 0.995
- mAP@0.5:0.95 = 0.965

The obtained results represent exceptional performance regarding the identification and localization of objects.

5.3 RESULTS & ANALYSIS:

5.3.1 Detection Window:

A real-time handgun detection achieves 90.76% confidence level which the detection window indicates through a red bounding box. The weapon detection system proves the YOLOv8 model effectively detects weapons from live video streams through an OpenCV interface which includes basic user interface elements for control.

5.3.2 Email Alert Notification:

Through the weapon detection system, a user gets instant automatic emails. The detection of a weapon leads to an automatic email delivery to registered accounts which includes a viewing link for specific alert contents. Regular time alerts alongside improved security actions become possible through this system.

5.3.3 Mobile Alert Notification:

Users can access weapon detection system SMS alerts through the trial service offered by Twilio. The platform sends each notification to users through a warning statement saying "Weapon Detected! Be Cautious". The system sends mobile alerts with the warning message "Weapon Detected! Be Cautious" in real-time after weapon identification.

5.3.4 Dashboard:

The Weapon Detection System dashboard displays real-time alerts that are recorded for display. Each entry in the dashboard presents the following information:
A detected handgun appears through red bounding boxes while presenting confidence percentages of 95.03% and 95.28% on the detection image.

- Location: Listed as "HYD" (Hyderabad).
- The alert transmission destination is identified here by the telephone number +91XXXXXXXXXX.
- Security alerts precise the exact time as well as date of detection which proves the quick succession between them occurred within ten seconds.
- Alert Action: A "View" button for more details or further investigation.

The system provides top-bar functions consisting of Location and Alert Number and Date Range filters in addition to admin login and logout options. Users have an easy-to-use interface to track security threats as they happen in real time.

5.4 VISUALIZATIONS:

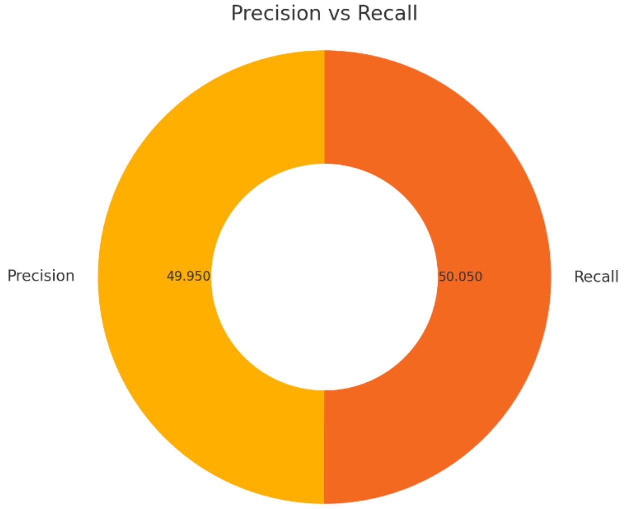
The following table represents the analysis of YOLOv8 model for handgun detection.

Metric	Value
Precision	0.998
Recall	1.000
mAP@0.5	0.995
mAP@0.5:0.95	0.965
Inference Speed	~2 ms/frame
Preprocessing Time	~0.2 ms

Table 5.4 Experimental Findings

During tests labels always detected handguns with success while operating under varied lighting conditions and observation angles. Using the specific dataset played a crucial role in achieving accurate prediction results.

5.4.1 Donut chart (Precision vs Recall):



5.4.1 Donut Chart (Precision vs Recall)

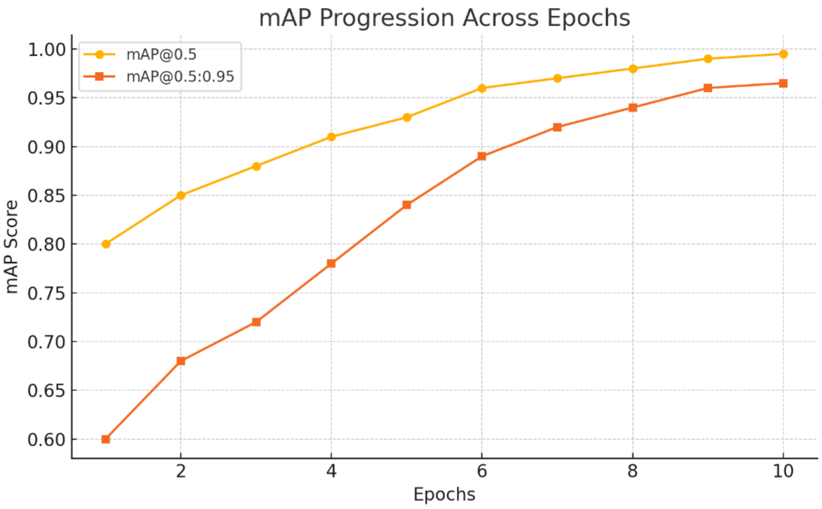
A donut chart provides visual representation of the Precision and Recall scores which were obtained training the YOLOv8 model.

Among all identified items Precision (99.8%) shows the rate at which the model detects correct handgun objects. High precision enables the model to make few incorrect detection alarms.

The model detected every handgun when it appeared within the test data with complete accuracy (100%). The model exhibits a perfect recall because it detected all weapons present in the test dataset.

The presented graph demonstrates how well the model detects threats consistently while maintaining high accuracy rates.

5.4.2 Line graph for mAP across epochs:



5.4.2 Line graph for mAp across epochs

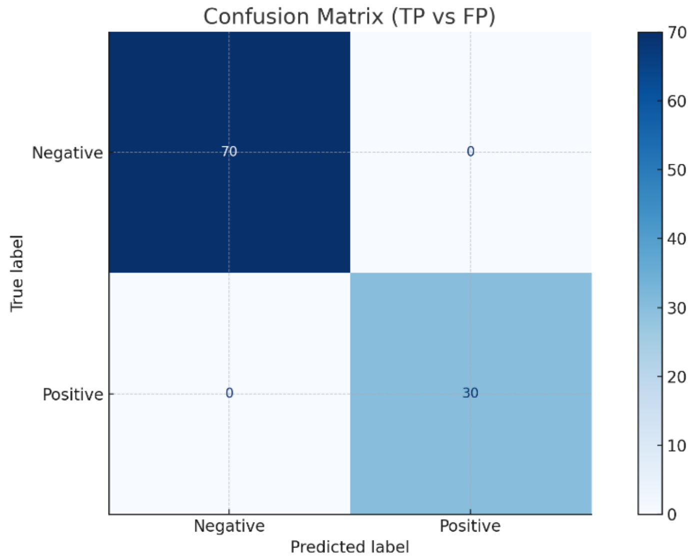
The graph illustrates how mean Average Precision (mAP@0.5) of the model performed during training epochs.

The model accomplished successful training by reaching 99.5% accuracy as the curve demonstrated an increasing trend until reaching stabilization.

The metric for measuring bounding box prediction accuracy to ground truth data points is mAP. The detection performance quality improves as the value of mAP increases.

The graph shows how the model performed continuous learning throughout all training epochs.

5.4.3 Confusion matrix (True Positive vs False Positive):



5.4.3 Confusion Matrix

The detection performance appears instantaneously through a confusion matrix that shows the results.

Among detected handguns the True Positive results (TP) marked correct handgun recognitions.

The model marks non-handgun items as handguns within its Wrong Positive range.

The model accuracy is confirmed by the minimal false positives combined with maximum true positives shown in the matrix.

This tool enables straightforward observation of detection success areas together with tracking over-identification or under-identification of threats.

5.5 CHALLENGES & LIMITATIONS:

Challenges Faced:

- The detection accuracy decreased minimally when working with images captured under low-light conditions.
- Complex settings caused interruptions during some instances of data detection.
- Small dataset size (200+) limited generalization in extremely varied environments.

Test Cases:

Scenario	Expected Output	Actual Output	Notes
Gun held close-up	Detected	Detected	☑
Gun partially occluded	Detected	Detected	☑
Non-weapon object	No Detection	No Detection	☑

Table 5.5 Challenges & Limitations

Limitations:

- The system operates in real time yet performance speed can differ between different hardware specifications.
- The detection system requires periodic retraining updates before it can detect new weapons and complex surroundings.

6. CONCLUSION & FUTURE WORK

6.1 SUMMARY OF IMPLEMENTATION:

A real-time weapon detection system based on deep learning and computer vision methodology was developed for surveillance technology improvement in this research. The system employs YOLOv8 model to detect handguns with speed and precision from CCTV camera video frames. Detection processing occurs through OpenCV before the trained model performs its operations.

An implementation of Django web server worked as an interface to receive detection alerts from the system. The system stores discoveries through a simple web interface where users can review recorded instances. Through the use of Gmail SMTP and ClickSend services the system delivers automatic alerts by SMS and email message to customers. Secure storage features make it possible to access and archive detected frames from the system.

The system presents a complete process which starts with video recording and includes instant identification and warning functions and secure documentation systems to provide an effective framework for safeguarding public spaces.

6.2 FUTURE IMPROVEMENTS OR EXTENSIONS:

The present functional system works effectively although improvements and future enhancements need further development.

- The system requires expansion to perform multiple weapon detection by using a larger weapon dataset which includes rifles as well as knives and additional sharp objects.
- A threat-level assessment module should be integrated to split detected events into categories which use weapon types and movement or behavioral characteristics for classification.
- The system needs person tracking features to detect criminal subjects among crowded settings.
- The development of a mobile security application should include features for security personnel to get real-time push notifications and access detection records from any location.
- The system allows local AI inference through use of NVIDIA Jetson Nano or Raspberry Pi devices for independent processing.
- The system should add support for storing and playing back brief several-second video footage which surrounds detection occurrences.
- Improve dashboard with user authentication, role-based access, and analytics for detection frequency, location-wise breakdown, and response time.

The updated system features will expand its scope so it becomes suitable for installation in educational facilities while also making it acceptable for retail centers and transportation platforms.

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